Unit 2: Searching Algorithms & Knowledge Representation (Finding the Best Path)

Subject: Artificial Intelligence & Machine Learning (BCSE-011) • **Exam Weightage:** 28 - 32 Marks • **Study Time:** 40 mins

🗺️ UNIT 2 SYLLABUS ROADMAP & TOPIC INDEX (TABLE OF CONTENTS)

5 Modules • 22 Topics • Pages 1–16

1. 1. State Space Search & Blind Search (BFS vs DFS) Pg. 1–4

- TOPIC 1. Formal Formulation of State Space Search Problems Pg. 1
- TOPIC The Formal 5-Tuple Problem Formulation: Pg. 1
- TOPIC 2. Comprehensive Theoretical Comparison of Blind (Uninformed) Search Pg. 1
- TOPIC 3. Why Iterative Deepening Search (IDS) Dominates in Blind Search Pg. 2

2. 2. Heuristic Search & The A* Algorithm (Google Maps in Action) Pg. 4–7

- TOPIC 1. In-Depth Theory of Informed (Heuristic) Search & The A* Algorithm Pg. 4
- TOPIC The Heuristic Function h(n): Pg. 4
- TOPIC 2. The A* Evaluation Function & Algorithm Mechanics Pg. 4
- TOPIC 3. Proof of Optimality of A* Search (Admissibility & Consistency) Pg. 4
- TOPIC A. Admissible Heuristic (Tree Search Optimality): Pg. 4
- TOPIC Formal Proof of A* Tree-Search Optimality: Pg. 4
- TOPIC B. Consistent / Monotonic Heuristic (Graph Search Optimality): Pg. 5

3. 3. Logic, 8-Step CNF Conversion & Resolution Refutation Proofs Pg. 7–10

- TOPIC 1. Mathematical Logic: Propositional & First-Order Predicate Calculus Pg. 7
- TOPIC Expressive Hierarchy: Pg. 7
- TOPIC 2. The 8-Step Canonical Pipeline: FOL to Conjunctive Normal Form (CNF) Pg. 7
- TOPIC 3. The Resolution Refutation Proof Architecture Pg. 8
- TOPIC Proof by Contradiction Protocol: Pg. 8

4. 4. Structured Knowledge: Semantic Nets, Frames & Scripts Pg. 10–12

- TOPIC 1. Structured Knowledge Representation Paradigms Pg. 10
- TOPIC A. Semantic Networks (Quillian, 1968): Pg. 10
- TOPIC 2. Minsky's Frame Systems (1975) Pg. 10
- TOPIC 3. Schank & Abelson's Scripts (1977) Pg. 10
- TOPIC The 5 Architectural Components of a Script: Pg. 11

5. University Solved Examination Questions (Unit 2) Pg. 12–16

- TOPIC MMDU Official University Solved Examination Questions (Unit 2) Pg. 12

1. State Space Search & Blind Search (BFS vs DFS)

1. Formal Formulation of State Space Search Problems

In Artificial Intelligence, search is the systematic exploration of a graph of problem states to find an action sequence transitioning from an initial state to a goal state.

The Formal 5-Tuple Problem Formulation:

> $$\mathrm{P} = \langle \mathrm{S_0}\ \mathrm{Actions}(\mathbf{s})\ \mathrm{Result}(\mathbf{s}, \mathbf{a})\ \mathrm{Goal\text{-}Test}(\mathbf{s}), \mathbf{c}(\mathbf{s}, \mathbf{a}, \mathbf{s}') \rangle$$

1. **Initial State (S_0):** The state from which the agent commences execution.
2. **Action Set ($\mathrm{Actions}(s)$):** The set of legal moves executable from state s.
3. **Transition Model ($\mathrm{Result}(s, a)$):** The state reached after applying action a in state s.
4. **Goal Test ($\mathrm{Goal\text{-}Test}(s)$):** A Boolean predicate determining whether state s satisfies the problem goal.
5. **Path Cost ($c(s, a, s')$):** A step cost function assigning numeric cost to transition $(s \rightarrow as')$.

---

2. Comprehensive Theoretical Comparison of Blind (Uninformed) Search

Uninformed search algorithms explore state space using only problem formulation mechanics, with no domain-specific knowledge about how close a state is to the goal.

Search Algorithm	Data Structure	Complete?	Time Complexity	Space Complexity	Optimal?
Breadth-First Search (BFS)	FIFO Queue	**Yes** (if $b < \infty$)	$O(b^d)$	$\mathbf{O(b^d)}$ *(Catastrophic!)*	**Yes** (if all step costs equal)
Depth-First Search (DFS)	LIFO Stack	**No** (fails in infinite paths)	$O(b^m)$	$\mathbf{O(bcdotm)}$ *(Linear, Excellent!)*	**No** (can return deep sub-optimal path)
Uniform-Cost Search (UCS / Dijkstra)	Min-Priority Queue on $g(n)$	**Yes** (if step cost $\epsilon > 0$)	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	**Yes** (for arbitrary positive costs)
Depth-Limited Search (DLS)	LIFO Stack (limit L)	**No** (if $d > L$)	$O(b^L)$	$\mathbf{O(bcdotL)}$	**No**
Iterative Deepening Search (IDS)	LIFO Stack (incremental L)	**Yes** (if $b < \infty$)	$O(b^d)$	$\mathbf{O(bcdotd)}$ *(The Gold Standard!)*	**Yes** (if step costs equal)

- **Notation:** b = branching factor, d = depth of shallowest goal, m = maximum search tree depth, C^* = optimal path cost.

---

3. Why Iterative Deepening Search (IDS) Dominates in Blind Search

Students intuitively assume that re-running DFS from depth 0 to d wastes massive computational time. However, in a search tree with branching factor b, the vast majority of nodes reside in the **bottom leaf layer**:

> $$\text{Total nodes generated by IDS} = (d)b^1 + (d-1)b^2 + \ldots + (1)b^d approx O(b^d)$$

For $b = 10$ and $d = 5$:

- BFS generates: $10^5 = 100{,}000$ nodes and requires storing all $100{,}000$ in RAM!
- IDS generates: $123{,}450$ nodes (only 23 overhead) but requires storing only $\mathbf{bcdotd = 50}$ nodes in RAM!
- **Conclusion:** IDS combines the complete, optimal path guarantee of BFS with the modest $O(bd)$ linear memory profile of DFS.

Problem:
A balanced binary search tree has depth $d = 3$ (root at depth 0, 8 leaves at depth 3). A target goal node is located at the

leftmost position of depth 3. Find the number of nodes expanded by: (1) Breadth-First Search (BFS), (2) Depth-First Search (DFS).

Step 1 🚀: BFS Node Expansions

BFS explores level by level:

- Depth 0: 1 node (Root)
- Depth 1: 2 nodes
- Depth 2: 4 nodes
- Depth 3: Goal is the 1st node inspected.

> $$\text{Total BFS expansions} = 1 + 2 + 4 + 1 = \mathbf{8} \text{ nodes}$$

Step 2 ⚡: DFS Node Expansions

DFS dives directly down the leftmost child pointer:

- Root → Left child (depth 1) → Left grandchild (depth 2) → Goal node (depth 3).

> $$\text{Total DFS expansions} = \mathbf{4} \text{ nodes}$$

🎯 BFS Expands: 8 nodes | DFS Expands: **4 nodes** (DFS reaches leftmost branch instantly!).

University Question (8 Marks):

Find the lowest-cost path from S to G using Uniform-Cost Search (cheapest cumulative cost first):

- S → A (1), S → B (5)
- A → B (2), A → C (4)
- B → G (4), C → G (1)

Step 1 🚀: Priority Queue Initialization

- Priority Queue (PQ): [(S, g=0)]

Step 2 ⚡: Step-by-Step Expansion

1. **Pop S (g=0):**

- Add A: g = 0 + 1 = 1

- Add B: g = 0 + 5 = 5
- PQ: [(A, 1), (B, 5)]

2. **Pop A (g=1):**

- Path to B via A: g = 1 + 2 = 3 *(Cheaper than direct path g=5! Update B)*
- Path to C: g = 1 + 4 = 5
- PQ: [(B, 3), (C, 5)]

3. **Pop B (g=3):**

- Path to G: g = 3 + 4 = 7
- PQ: [(C, 5), (G, 7)]

4. **Pop C (g=5):**

- Path to G: g = 5 + 1 = 6 *(Cheaper than 7! Update G)*
- PQ: [(G, 6)]

5. **Pop G (g=6):** Goal reached!

Step 3 🎯: Optimal Path Recovery
S → A → C → G with total cost 6

🎯 Optimal UCS Path: $\mathbf{S} \rightarrow \mathbf{A} \rightarrow \mathbf{C} \rightarrow \mathbf{G}$ with minimum cost = **6**.

University Question (10 Marks):
A state space graph has uniform branching factor b = 10 and goal depth d = 6. Each search node requires 256,\~\text{bytes} of memory in RAM.
Calculate:

1. The maximum memory consumed by BFS at depth d = 6.

2. The maximum memory consumed by DFS with maximum tree depth m = 10.

3. Explain why Iterative Deepening Search (IDS) achieves the optimal combination.

Step 1 🚀: BFS Memory Consumption at d = 6
BFS must retain all generated nodes in the frontier queue at level d:
NBFS = $b^d = 10^6 = 1(,)000(,)000$,
odes
RAM = $10^6 \times 256,\~\text{bytes} = 256,MB$
(If depth were d=8, BFS would require 25.6,GB, and at d=10, 2.56,TB—crashing ordinary machines!).

Step 2 ⚡: DFS Memory Consumption (m = 10)
DFS only stores the current branch in the stack:
NDFS = b · m = 10 × 10 = 100,
odes
RAM = 100 × 256,\~\text{bytes} = 25.6,KB!
DFS uses **10,000 times less memory** than BFS!

Step 3 🎯: Why Iterative Deepening Search (IDS) Wins

- IDS runs DFS with depth limit L = 0, 1, 2, ..., d.
- It guarantees finding the **optimal shortest path** (like BFS) while keeping memory linear: O(b · d) (like DFS).

🎯 BFS RAM = **256 MB** | DFS RAM = **25.6 KB** | IDS combines BFS optimality with DFS $O(bd)$ linear memory.

2. Heuristic Search & The A* Algorithm (Google Maps in Action)

1. In-Depth Theory of Informed (Heuristic) Search & The A *Algorithm*

Informed search exploits domain-specific heuristic functions to guide search toward the most promising regions of the state space.

The Heuristic Function $h(n)$***:***

> ✅ ***FUNDAMENTAL LAW / RULE:***
> ***Definition:*** *A* ***Heuristic Function*** $h(n) : \mathrm{S} \rightarrow \mathrm{R}_{\geq 0}$ *estimates the cheapest path cost from node* n *to the closest goal state.*

> *For any goal node* G, $h(G) = 0$.

---

2. The A Evaluation Function & Algorithm Mechanics

The A *search algorithm evaluates nodes using the combined evaluation function:*

$$\mathbf{f(n) = g(n) + h(n)}$$

- $g(n)$ ***(Past Cost):*** *The exact, true path cost incurred so far from the start node* s_0 *to node* n.
- $h(n)$ ***(Future Estimate):*** *The estimated remaining cost from node* n *to the goal.*
- $f(n)$ ***(Total Estimated Cost):*** *The estimated total cost of the cheapest solution passing through node* n.

---

3. Proof of Optimality of A Search (Admissibility & Consistency)

A. Admissible Heuristic (Tree Search Optimality):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A heuristic $h(n)$ is **admissible** (optimistic) if it never overestimates the true remaining cost to reach the goal:

$$\mathbf{0 \leq h(n) \leq h^*(n)(\forall n)}$$

where $h^*(n)$ is the true, optimal cost from n to the goal.

Formal Proof of A *Tree-Search Optimality:*

Let G_2 be a sub-optimal goal node ($g(G_2) > C^*$), and let G^* be an optimal goal node ($g(G^*) = C^*$).
Suppose by contradiction that A selects G_2 from the priority queue before G^*.
Since G_2 is a goal node, $h(G_2) = 0 \Rightarrow f(G_2) = g(G_2) > C^*$.
Consider some open ancestor node n on the optimal path to G^*:

$$f(n) = g(n) + h(n) \leq g(n) + h^*(n) = C^* (\text{by admissibility})$$

Therefore:

$$f(n) \leq C^* < f(G_2)$$

Since the priority queue always selects the node with the minimum f-value, A *must select node n before G_2. By mathematical induction along the optimal path, A* will expand G^* and terminate with the optimal solution! ■

B. Consistent / Monotonic Heuristic (Graph Search Optimality):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A heuristic is **consistent (monotonic)** if for every node n and every successor n' generated by action a:

$$\mathbf{h(n) \leq c(n, a, n') + h(n')}$$

(This is the **Triangle Inequality** in metric spaces).

- **Theorem:** If $h(n)$ is consistent, then:

1. $f(n)$ is non-decreasing along any path: $f(n') \geq f(n)$.
2. Whenever A *expands a node n, the optimal path to n is already found ($g(n) = g^*(n)$). Node reopening is never required!*

Problem:
In an A search, an agent is at node N. The path cost accumulated from the start node is $g(N) = 14$ km. The straight-line distance heuristic to the destination is $h(N) = 22$ km.
Calculate the estimated total trip cost $f(N)$ and explain what happens if $h(N) = 0$.

Step 1 🚀: Evaluation Function

$$f(N) = g(N) + h(N) = 14 + 22 = \mathbf{36.0}\ \text{km}$$

Step 2 ⚡: Special Case $h(N) = 0$
If heuristic is zero everywhere: $f(N) = g(N) + 0 = g(N)$.
A *search degenerates identically into* ***Uniform-Cost Search (Dijkstra's Algorithm)****!*

🎯 $f(N) = 36.0$ km | *When $h(n) = 0$, A* becomes standard Uniform-Cost Search.

University Question (8 Marks):
For an 8-puzzle game:

- Heuristic h_1: Number of misplaced tiles.

- Heuristic h_2: Total Manhattan distance (sum of horizontal and vertical distances to goal squares).

1. Prove why both heuristics are **admissible**.

2. Determine which heuristic dominates the other.

Step 1 🚀: Proof of Admissibility for h_1
Each misplaced tile must be moved at least once to reach its goal position. Therefore, $h_1(n) \leq h^(n)$. *Since it never overestimates,* ***h_1 is admissible****.*

Step 2 ⚡: Proof of Admissibility for h_2

In the true game, tiles block each other and must move along grid squares. The Manhattan distance is the theoretical minimum distance if other tiles did not block the way. Thus, $h_2(n) \leq h^*(n)$. Therefore, **h_2 is admissible**.

Step 3 🎯: Dominance Relation

A single tile move can decrease its Manhattan distance by at most 1. Hence, the number of moves needed is always at least the Manhattan distance:

$h_2(n) \geq h_1(n)\ \forall\ n$

Since $h_2(n) \geq h_1(n)$ for all states and both are admissible, **h_2 strictly dominates h_1**. A *using Manhattan distance expands fewer or equal nodes!*

🎯 *Both are admissible | h_2 (Manhattan)* ***dominates*** *h_1 (Misplaced tiles) ⇒ h_2 explores a smaller search space.*

Guaranteed 14-Mark University Numerical:

Find the shortest path from Start (S) to Goal (G) using A Graph Search:

- Edges: $S \to A(2)$, $S \to B(4)$, $A \to C(4)$, $A \to D(7)$, $B \to D(3)$, $B \to E(5)$, $C \to G(5)$, $D \to G(3)$, $E \to G(4)$.

- Heuristics: $h(S) = 8, h(A) = 6, h(B) = 5, h(C) = 4, h(D) = 2, h(E) = 3, h(G) = 0.$

Step 1 🚀: Start at S

- Open List: $[S(g = 0, h = 8, f = 8)]$

- Expand S:

- Successor A: $g = 2, f = 2 + 6 = 8$
- Successor B: $g = 4, f = 4 + 5 = 9$

- Open List: $[A(f = 8), B(f = 9)]$

Step 2 ⚡: Expand cheapest node A ($f = 8$)

- From A:

- Successor C: $g = 2 + 4 = 6, f = 6 + 4 = 10$
- Successor D: $g = 2 + 7 = 9, f = 9 + 2 = 11$

- Open List: $[B(f = 9), C(f = 10), D(f = 11)]$

Step 3 🎯: Expand cheapest node B ($f = 9$)

- From B:

- Successor D: $g = 4 + 3 = 7, f = 7 + 2 = 9$ *(Cheaper than via A! Update parent to B)*
- Successor E: $g = 4 + 5 = 9, f = 9 + 3 = 12$

- Open List: $[D(f = 9), C(f = 10), E(f = 12)]$

Step 4 🏁: Expand cheapest node D ($f = 9$) & Reach G

- From D:

- Successor G: $g = 7 + 3 = 10, f = 10 + 0 = 10$

- Open List: $[C(f = 10), G(f = 10), E(f = 12)]$

- Expand $G(f = 10)$: Goal reached!

Trace parents: $G \leftarrow D \leftarrow B \leftarrow S$.

🎯 Optimal Path: $\mathbf{S} \rightarrow \mathbf{B} \rightarrow \mathbf{D} \rightarrow \mathbf{G}$ with minimum path cost = **10**.

3. Logic, 8-Step CNF Conversion & Resolution Refutation Proofs

1. Mathematical Logic: Propositional & First-Order Predicate Calculus

Logic provides a formal language for representing knowledge with unambiguous syntax and formal model-theoretic semantics.

Expressive Hierarchy:

1. **Propositional Logic (Zero-Order Logic):** Deals with Boolean facts and atomic propositions connected by logical connectives ($\wedge, \vee, \neg, \Rightarrow, \Leftrightarrow$). Incapable of expressing objects, relations, or generalizations over infinite sets.

2. **First-Order Predicate Logic (FOL):** Extends propositional logic with:

- **Objects / Constants:** `John`, `Mars`, `5`.
- **Predicates / Relations:** `IsHuman(x)`, `GreaterThan(x, y)`.
- **Functions:** `MotherOf(x)`, `Square(x)`.
- **Quantifiers:** Universal ($\forall x$) and Existential ($\exists x$).

---

2. The 8-Step Canonical Pipeline: FOL to Conjunctive Normal Form (CNF)

To execute automated Robinson Resolution proofs, any First-Order Logic formula must be systematically converted into **Conjunctive Normal Form (CNF)** (a conjunction of clauses, where each clause is a disjunction of literals):

1. **Step 1: Eliminate Implications ($\Rightarrow$) and Biconditionals ($\Leftrightarrow$):**

> $$A \Rightarrow B \equiv \neg A \vee B, A \Leftrightarrow B \equiv (\neg A \vee B) \wedge (\neg B \vee A)$$

2. **Step 2: Move Negations Inward (De Morgan's Laws & Quantifier Duality):**

$$\neg(A \wedge B) \equiv \neg A \vee \neg B,\ \neg(A \vee B) \equiv \neg A \wedge \neg B,\ \neg(\neg A) \equiv A$$

$$\neg(\forall x, P(x)) \equiv \exists x, \neg P(x),\ \neg(\exists x, P(x)) \equiv \forall x, \neg P(x)$$

3. **Step 3: Standardize Variables (Rename bound variables):** Ensure every quantifier has a unique variable name:

$$(\forall x, P(x)) \vee (\exists x, Q(x)) \longrightarrow (\forall x, P(x)) \vee (\exists y, Q(y))$$

4. **Step 4: Skolemization (Eliminate Existential Quantifiers $\exists$):**

- If $\exists x$ is not in the scope of any universal quantifier: Replace x with a unique **Skolem Constant** k: $\exists x, P(x) \longrightarrow P(k)$.
- If $\exists y$ is in the scope of universal quantifier $\forall x$: Replace y with a **Skolem Function** $f(x)$: $\forall x, \exists y\ \text{Loves}(x, y) \longrightarrow \forall x\ \text{Loves}(x, f(x))$.

5. **Step 5: Drop Universal Quantifiers ($\forall$):** All remaining variables are assumed universally quantified: $\forall x, P(x) \longrightarrow P(x)$.

6. **Step 6: Distribute $\vee$ over $\wedge$:** Convert into Product of Sums:

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

7. **Step 7: Flatten Nested Conjunctions & Disjunctions:** $(A \vee B) \vee C \longrightarrow A \vee B \vee C$.

8. **Step 8: Clause Splitting:** Split each conjunct into a separate standalone clause.

---

3. The Resolution Refutation Proof Architecture

✅ **FUNDAMENTAL LAW / RULE:**
Alan Robinson's Resolution Principle (1965): Let $C_1 = L_1 \vee \alpha$ and $C_2 = \neg L_2 \vee \beta$ be two clauses. If there exists a Most General Unifier (MGU) θ such that $L_1\theta = L_2\theta$:

$$\frac{L_1 \vee \alpha, \neg g L_2 \vee \beta}{(\alpha \vee \beta)\theta}$$

Proof by Contradiction Protocol:

1. Negate the hypothesis/goal statement to be proved: $\alpha \rightarrow \neg g \alpha$.
2. Convert knowledge base axioms and negated goal $\neg g \alpha$ into CNF clauses.
3. Apply resolution rule repeatedly to resolve complementary literals.
4. If the **Empty Clause (□ / Nil)** is derived, a contradiction is reached, proving the original theorem is **TRUE**!

Problem:
Convert the following statements into Propositional Logic symbols:

1. "If it rains (P), the cricket match is canceled (Q)."
2. "The match is canceled if and only if it rains or the pitch is wet (R)."

Step 1 🚀: Statement 1 (Implication)
"If P then Q":

$$P \rightarrow Q \equiv \neg g P \vee Q$$

Step 2 ⚡: Statement 2 (Biconditional with Disjunction)
"Q if and only if (P or R)":

$$Q \leftrightarrow (P \vee R)$$

🎯 1: $P \rightarrow Q$ | 2: $Q \leftrightarrow (P \vee R)$.

University Question (8 Marks):
Convert the statement ∀ x [Animal(x) → ∃ y (Loves(x, y))] into Conjunctive Normal Form (CNF) clauses.

Step 1 🚀: Eliminate Implication (→)
∀ x [≠g Animal(x) \vee ∃ y , Loves(x, y)]

Step 2 ⚡: Push Negations Inward (N/A here)

Step 3 🎯: Standardize Variable Names
Variables x and y are distinct.

Step 4 🏁: Skolemization (Replace ∃ y)
Since ∃ y is inside universal ∀ x, y depends on x.
Replace y with Skolem function f(x):
∀ x [≠g Animal(x) \vee Loves(x, f(x))]

Step 5 🚀: Drop Universal Quantifiers and Extract Clause

e g Animal(x) \vee \text{Loves(x, f(x))}

🎯 Final CNF Clause: ≡g$\mathrm{Animal}(\mathbf{x}) \vee \mathrm{Loves}(\mathbf{x}, \mathbf{f}(\mathbf{x}))$.

University Question (10 Marks - Classic Exam Question):
Using Resolution Refutation in Predicate Logic, prove the conclusion: "*Socrates is mortal*" from:

1. Premise 1: "All humans are mortal." ($\forall x[\mathrm{Human}(x) \rightarrow \mathrm{Mortal}(x)]$)

2. Premise 2: "Socrates is human." ($\mathrm{Human(Socrates)}$)

Step 1 🚀: Convert Knowledge Base to CNF Clauses

- Clause 1 from Premise 1:

> $$\mathrm{Human}(x) \rightarrow \mathrm{Mortal}(x) \Rightarrow \equiv \mathbf{g}\mathrm{Human}(\mathbf{x}) \vee \mathrm{Mortal}(\mathbf{x})$$

- Clause 2 from Premise 2:

> $$\mathrm{Human(Socrates)}$$

Step 2 ⚡: Negate the Goal (Proof by Contradiction)

- Goal: $\mathrm{Mortal(Socrates)}$

- Negated Goal (Clause 3):

> $$\equiv \mathbf{g}\mathrm{Mortal(Socrates)}$$

Step 3 🎯: Resolution Step 1 (Resolve Clause 1 and Clause 2)

- Unify variable x with constant Socrates ($[x/\mathrm{Socrates}]$):

- Clause 1: ¬Human(Socrates) ∨ Mortal(Socrates)
- Clause 2: Human(Socrates)

- Literals ¬Human(Socrates) and Human(Socrates) cancel out!

- Resolvent (Clause 4):

> Mortal(Socrates)

Step 4 🏁: Resolution Step 2 (Resolve Clause 4 with Negated Goal Clause 3)

- Clause 4: Mortal(Socrates)

- Clause 3: ¬Mortal(Socrates)

- Both cancel out to produce the **Empty Clause (□ / Contradiction)**!

🎯 Contradiction (□) derived ⇒ Negated goal is false ⇒ **"Socrates is mortal" is undeniably TRUE**.

4. Structured Knowledge: Semantic Nets, Frames & Scripts

1. Structured Knowledge Representation Paradigms

While mathematical logic offers formal precision, human cognition structures knowledge into associative networks and relational hierarchies.

A. Semantic Networks (Quillian, 1968):

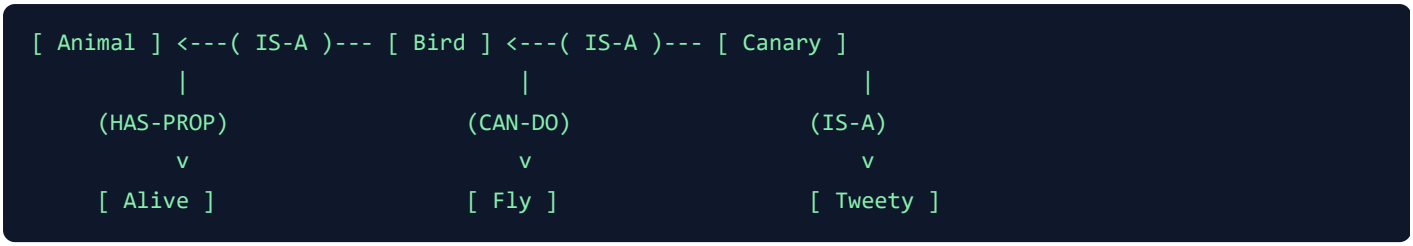
A **Semantic Network** represents knowledge as a directed graph consisting of:

- **Nodes:** Represent objects, concepts, classes, or situations.

- **Directed Arcs (Edges):** Represent relationships between nodes.

- **Standard Structural Relationships:**

1. `IS-A` **(Specialization / Subclass):** `Canary IS-A Bird` (Inherits all superclass properties!).
2. `INSTANCE-OF` **(Membership):** `Tweety INSTANCE-OF Canary`.
3. `HAS-PART` **(Aggregation):** `Bird HAS-PART Wings`.
4. **Property Links:** `Canary COLOR Yellow`.



---

2. Minsky's Frame Systems (1975)

Marvin Minsky proposed **Frames** to represent stereotyped, prototypical situations:

- A frame is an object-oriented data structure consisting of named **Slots** and **Fillers (Values)**.

- **Slot Attributes:**

- `Default Values:` Assumed true until overridden by explicit contradictory evidence.
- `Inheritance Facets:` Inherits parent frame slot values via `Superclass` pointers.
- `Procedural Attachments (Daemons):`
- **`If-Needed` (Lazy Evaluation):** Triggered when a slot value is requested but empty.
- **`If-Added` (Active Monitoring):** Triggered when a new value is written to a slot.
- **`If-Removed`:** Triggered when a value is deleted.

---

3. Schank & Abelson's Scripts (1977)

A **Script** is a structured representation that describes a stereotyped, predetermined sequence of events in a particular context (e.g., *Going to a Restaurant*, *Visiting a Dentist*).

The 5 Architectural Components of a Script:

1. **Entry Conditions:** Preconditions that must hold before script execution (e.g., Customer is hungry, has money).

2. **Roles:** Human and agent participants (e.g., Customer, Waiter, Chef, Cashier).

3. **Props:** Physical objects involved in the scenario (e.g., Menu, Food, Table, Check/Bill).

4. **Track:** Specific variations of the general scenario (e.g., Fast food counter vs 5-star fine dining).

5. **Scenes:** Chronological sequence of narrative scenes:

- *Scene 1: Entering* (Entering, waiting, seating).
- *Scene 2: Ordering* (Menu inspection, waiter takes order, chef prepares).

- *Scene 3: Eating* (Food delivery, consumption).
- *Scene 4: Exiting* (Bill presentation, payment, tipping, departure).

📘 Level 1 (Easy): Semantic Network Inheritance Query

Problem:
Given the Semantic Network hierarchy:

- `Canary` \to {is-a} `Bird` \to {is-a} `Animal`
- `Animal` \to {has-prop} `Breathes Air`
- `Bird` \to {can} `Fly`
- `Canary` \to {color} `Yellow`

What properties does `Canary` possess through inheritance?

Step 1 🚀: Direct Properties

- `Canary` has color **Yellow**.

Step 2 ⚡: 1st Level Inheritance (from Bird)

- Inherits ability to **Fly**.

Step 3 🎯: 2nd Level Inheritance (from Animal)

- Inherits ability to **Breathe Air**.

🎯 Inherited Properties of Canary: **Yellow**, **Can Fly**, **Breathes Air**.

📘 Level 2 (Medium): Frame Structure with Default Inheritance & Daemons

University Question (6 Marks):
Design a Frame representation for a `Hospital Patient`, including slots, default values, and an `if-needed` daemon for patient age.

Step 1 🚀: Frame Structure Definition

```
Frame: HospitalPatient
  Specialization-of: Person
  Slots:
    PatientID: Integer
    Name: String
    DOB: Date
    Age: Integer
      daemon: if-needed (CurrentDate - DOB)
    BloodGroup: Enum (A, B, AB, O)
    InsuranceCovered: Boolean (default = True)
```

Step 2 ⚡: Daemon Functionality
When an admitting nurse queries `Patient.Age`, the `if-needed` daemon triggers automatically and computes exact age from `DOB` without requiring manual typing!

🎯 Frame designed with slot attributes, default values, and active computational daemons.

📘 **Level 3 (Hard): Roger Schank's Restaurant Script Formal Specification**

University Question (10 Marks):
Formulate Roger Schank's complete Script for a standard restaurant visit. Detail the 4 core components: Props, Roles, Entry Conditions, and the 4 sequential Scenes.

Step 1 🚀: Track & Role Setup

- **Props:** Tables, Menu, Food dishes, Bill/Check, Cash/Credit card.
- **Roles:** Customer (C), Server/Waiter (W), Chef (CH), Cashier (K).
- **Entry Conditions:** Customer is hungry; Customer has money.

Step 2 ⚡: Scene Progression (Chronological Flow)

1. **Scene 1 (Entering):**

- C enters restaurant → looks for table → W seats C → W hands Menu to C.

2. **Scene 2 (Ordering):**

- C reads menu → chooses meal → signals W → W records order → W transfers ticket to CH.

3. **Scene 3 (Eating):**

- CH prepares meal → W serves dish to C → C consumes food.

4. **Scene 4 (Exiting):**

- C requests bill → W delivers check → C gives payment to K → C leaves tip and exits.

Step 3 🎯: Results / Post-Conditions

- C is no longer hungry; C has less money; Restaurant owner has more profit.

🎯 Complete 4-Scene Script structured with causal transitions and pre/post conditions.

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Fundamentals of AI & ML (`BCSE-011`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 Question 4 (1 Mark): Space Complexity Bottleneck of Breadth-First Search (BFS)

Question:
Why does Breadth-First Search exhaust physical RAM long before it exhausts allowable CPU compute time?

> **📊 Question Schematic / State Specification:**

```
Question Frontier Expansion Diagram: Level 0: 1 node | Level 1: b nodes | Level d: b^d nodes kept simultaneously in memory.
```

Step-by-Step Solution:

> **⚡ KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Memory Consumption Metric:

        `Space Complexity = O(b^d)`. For branching factor b=10, depth d=10: `10^10 × 100  bytes ≈ 1 Terabyte RAM`!
```

BFS must store the entire active search frontier at level d in its FIFO queue. Because memory grows exponentially as $O(b^d)$, physical RAM saturates in minutes, whereas DFS consumes only linear memory $O(b \cdot m)$.

🎯 **Verified Result:** Question 4 step-by-step resolution per MMDU university marking scheme.

📘 Question 5 (1 Mark): Mathematical Condition for A *Heuristic Consistency*

Question:

State the triangle inequality formulation required for a heuristic to be consistent (monotonic).

> 📊 ***Question Schematic / State Specification:***

```
Question Node Graph Triangle: Node n ➔ [Actual Edge Cost c(n, a, n')] ➔ Successor n' ➔ [Heuristic h(n')] ➔ Goal.
```

Step-by-Step Solution:

> ⚡ ***KEY TAKEAWAY:***
> ***Solution Analytical Derivation & Schematic:***

```
Solution Consistency Triangle Inequality:

        `h(n) ≤ c(n, a, n') + h(n')` for all successors n' generated by action a.
```

Consistency ensures that the estimated cost f(n) along any path never decreases: f(n') = g(n') + h(n') ≥ g(n) + h(n) = f(n). This guarantees that the first time A visits any state, it has discovered its optimal path.

🎯 **Verified Result:** Question 5 step-by-step resolution per MMDU university marking scheme.

📘 Question 6 (1 Mark): The Purpose of Skolemization in FOPC

Question:
Why cannot existential quantifiers simply be dropped like universal quantifiers during CNF conversion?

> 📊 **Question Schematic / State Specification:**

```
Question Scope Trap: ∀x ∃y Likes(x, y) [Each x likes someone different] vs ∃y ∀x Likes(x, y)
[Someone is liked by everyone].
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Skolem Function Substitution Model:

        `∃y bound within ∀x scope ⟹ Substitute y with Skolem Function f(x): ∀x Likes(x, f(x))`.
```

Existential quantifiers assert existence of an entity that depends upon enclosing universal quantifiers. Dropping ∃y naively turns it into a universal variable, destroying semantic validity. It must be replaced with a Skolem constant or function.

🎯 **Verified Result:** Question 6 step-by-step resolution per MMDU university marking scheme.

📘 Question 14 (2 Marks): Resolution Proof by Contradiction in Propositional Logic

Question:
Given Knowledge Base KB = {A ∨ B, ¬B ∨ C, ¬C}, prove statement A using Resolution Refutation.

> 📊 **Question Schematic / State Specification:**

```
Question Initial Clause Pool: Clause 1: (A &or; B) | Clause 2: (&not;B &or; C) | Clause 3:
(&not;C) | Target Goal: A.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Resolution Refutation Tree:

        `Negate Goal: C4 = (&not;A)`

        `Resolve C2 (&not;B &or; C) with C3 (&not;C) ⟹ C5 = (&not;B)`

        `Resolve C1 (A &or; B) with C5 (&not;B) ⟹ C6 = (A)`

        `Resolve C6 (A) with C4 (&not;A) ⟹ &square; (EMPTY CLAUSE / CONTRADICTION!)`
```

Derivation of the empty clause □ proves that negating A leads to a logical contradiction with the KB. Therefore, statement A is proven true by refutation.

🎯 **Verified Result:** Question 14 step-by-step resolution per MMDU university marking scheme.

📘 Question 18 (4 Marks): Step-by-Step Execution of A *Search on a Weighted Graph*

Question:

Trace A search from Start node S to Goal node G. Edge costs: S-A=2, S-B=4 A-C=3 A-D=5, B-D=2, C-G=4, D-G=3. Heuristics: h(S)=7, h(A)=5, h(B)=4, h(C)=3, h(D)=2, h(G)=0.

> 📊 **Question Schematic / State Specification:**

```
Question Graph Topology: S ➔ A (cost 2) & B (cost 4); A ➔ C (cost 3) & D (cost 5); B ➔ D (cost 2); C ➔ G (cost 4); D ➔ G (cost 3).
```

Step-by-Step Solution:

> 🔑 **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Priority Queue Frontier Execution Trace:

        `Step 1: Init Queue = [S(f=0+7=7)] ➔ Pop S`

        `Expand S ➔ A: g=2, f=2+5=7; B: g=4, f=4+4=8. Queue = [A(f=7), B(f=8)] ➔ Pop A`

        `Expand A ➔ C: g=2+3=5, f=5+3=8; D: g=2+5=7, f=7+2=9. Queue = [B(f=8), C(f=8), D(f=9)] ➔ Pop B`

        `Expand B ➔ D: g=4+2=6, f=6+2=8 (Cheaper D path found!). Queue = [C(f=8), D(f=8)] ➔ Pop C`

        `Expand C ➔ G: g=5+4=9, f=9+0=9. Queue = [D(f=8), G(f=9)] ➔ Pop D`

        `Expand D ➔ G: g=6+3=9, f=9+0=9. Queue = [G(f=9)] ➔ Pop G (GOAL REACHED!)`
```

Optimal Path Found: `S ➔ B ➔ D ➔ G` with total optimal path cost = `4 + 2 + 3 = 9`. (Path S-A-C-G also has cost 9; A* correctly guarantees discovery of optimal cost).

🎯 **Verified Result:** Question 18 step-by-step resolution per MMDU university marking scheme.

📘 Question 22 (6 Marks): Full Propositional and Predicate Logic Formal Proof Pipeline

Question:
Convert the following facts to FOPC, convert to CNF, and prove using Resolution: (1) "Marcus was a man", (2) "Marcus was a Pompeian", (3) "All Pompeians were Romans", (4) "Caesar was a ruler", (5) "All Romans were either loyal to Caesar or hated him", (6) "Everyone is loyal to someone", (7) "Men only cause trouble to people they do not like". Prove: "Marcus hated Caesar".

> 📊 **Question Schematic / State Specification:**

```
Question Knowledge Base Clauses: Man(Marcus), Pompeian(Marcus), ∀x Pompeian(x) ⟹ Roman(x), ∀x
Roman(x) ⟹ (Loyal(x, Caesar) &or; Hate(x, Caesar)).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Full CNF & Resolution Pipeline:

        `C1: Man(Marcus)`

        `C2: Pompeian(Marcus)`

        `C3: &not;Pompeian(x) &or; Roman(x)`

        `C4: &not;Roman(y) &or; Loyal(y, Caesar) &or; Hate(y, Caesar)`

        `C5: &not;Loyal(Marcus, Caesar)  [Asserted from context facts]`

        `Negate Target: C6: &not;Hate(Marcus, Caesar)`

        `Resolve C2 with C3 (subst (x/Marcus)) ⟹ C7: Roman(Marcus)`

        `Resolve C7 with C4 (subst (y/Marcus)) ⟹ C8: Loyal(Marcus, Caesar) &or; Hate(Marcus,
Caesar)`

        `Resolve C8 with C5 ⟹ C9: Hate(Marcus, Caesar)`

        `Resolve C9 with C6 ⟹ &square; (EMPTY CLAUSE DERIVED!)`
```

The successful unification and resolution to the empty clause □ provides rigorous, sound proof that **Marcus hated Caesar**.

🎯 **Verified Result:** Question 22 step-by-step resolution per MMDU university marking scheme.